

```

%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;

% retrieve selected cuts used for further fitting
[GH_FFCutsFits,CutFitFileGH] =
recover_selected_cuts('GH_FFCutsFits','e:\SHARE\FE\Data\analysis\07_fit_with_multicu
t_all\sel_cuts');

```

```

fn = fieldnames(GH_FFCutsFits);
%nf = 11
%for nf = 2:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei800_off110GHminFF';
if ismember(selectCuts(1:3),{'msk','dat'})||...
    contains(selectCuts,'all_fit_par')
    %continue
    error('Can not fit field: %s',selectCuts);
end

fprintf('*****\n');

```

```

*****

```

```

fprintf('Processing cuts: %s \n', selectCuts);

```

```

Processing cuts: Ei800_off110GHminFF

```

```

fprintf('*****\n');

```

```

*****

```

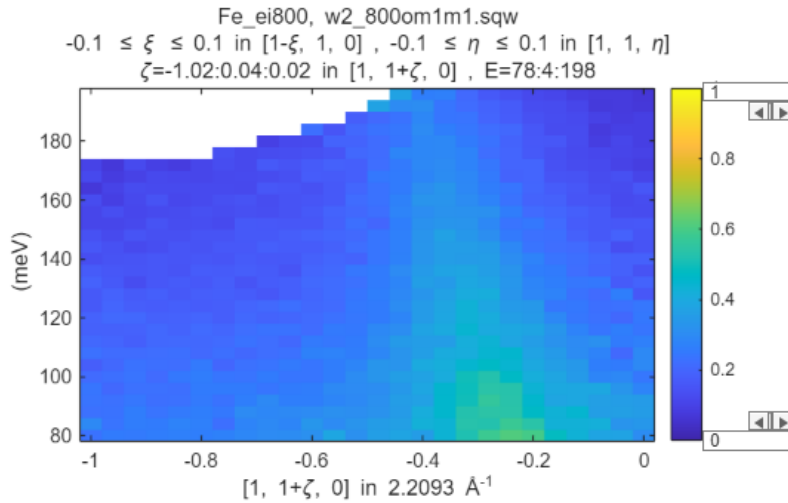
```

cuts_list = GH_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts,'Ei200')
    min_Ei = 30;
    mi = maps_instrument(200,600,'S');
end
if contains(selectCuts,'Ei400')
    min_Ei = 40;
    mi = maps_instrument(401,600,'S');
end
if contains(selectCuts,'Ei800')
    min_Ei = 70;
    mi = maps_instrument(786,600,'S');
end
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for j=1:numel(cuts_list)

```

```
cuts_list{j} = cuts_list{j}.set_instrument(mi);
cuts_list{j} = cuts_list{j}.set_sample(sample);
plot(cuts_list{j}); lz 0 1; keep_figure;
```

end



```
dir_name = "GH";

dE_step = 2; %original energy transfer step data were binned to. No point in going
finer
half_dE = 5; % half width of data binning

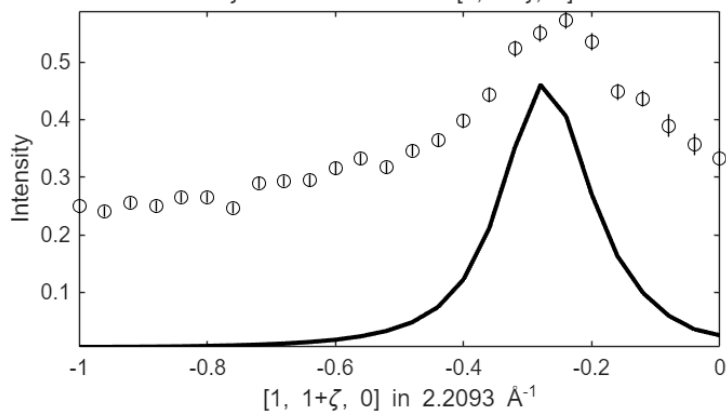
en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(min_Ei,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 170;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
GH_FFCutsFits.(selectCuts) = cuts_list;

GH_FFCutsFits = fit_multicutsFF_along_direction(...
    GH_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

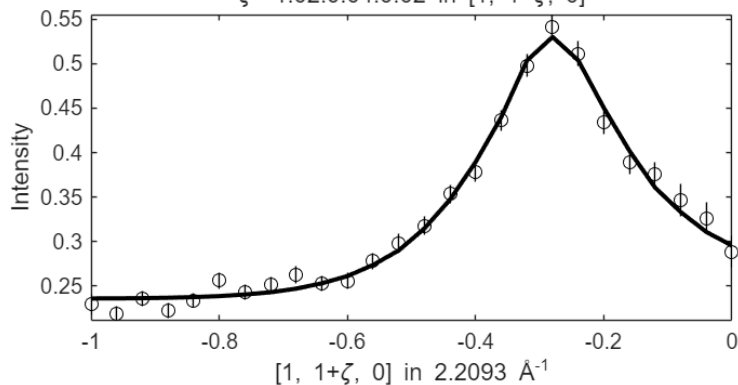
```
*****
**** En = 83±5 going to 193
*****
```

Fe_ei800, w2_800om1m1.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $78 \leq E \leq 88$ eV
 $\zeta = -1.02:0.04:0.02$ in $[1, 1+\zeta, 0]$

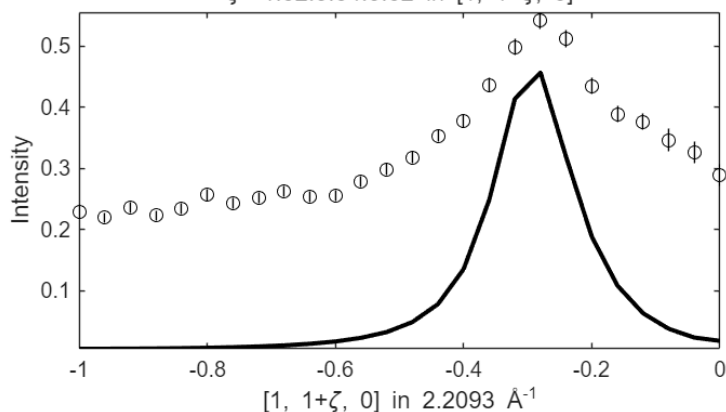


**** En = 93±5 going to 193

Fe_ei800, w2_800om1m1.sqw
 $S=0.95 \pm 0.1$; $J_0=32 \pm 1.5$; $J_1=0 \pm 0$; $\gamma=50.1625 \pm 9.38389$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $88 \leq E \leq 98$ eV
 $\zeta = -1.02:0.04:0.02$ in $[1, 1+\zeta, 0]$

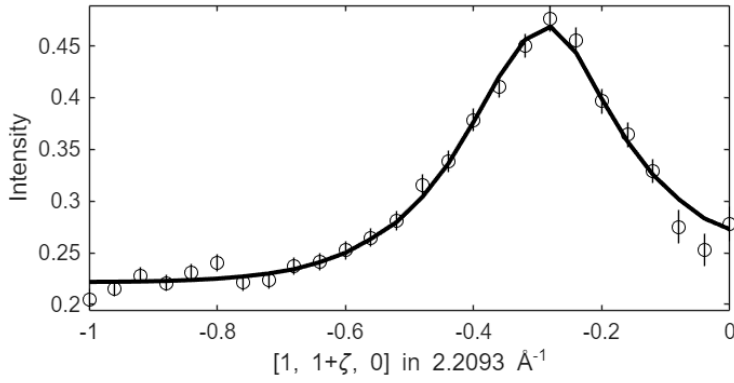


Fe_ei800, w2_800om1m1.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $88 \leq E \leq 98$ eV
 $\zeta = -1.02:0.04:0.02$ in $[1, 1+\zeta, 0]$

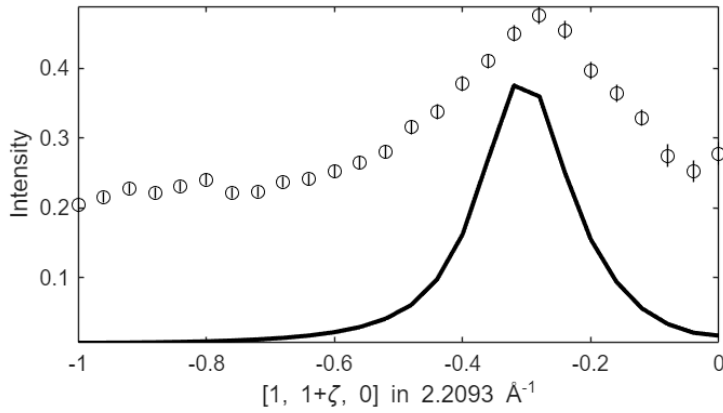


**** En = 103±5 going to 193

Fe_ei800, w2_800om1m1.sqw
 $S=0.98\pm0.019$; $J_0=34\pm0.49$; $J_1=0\pm0$; $\gamma=65.6312\pm1.3296$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $98 \leq E \leq 1$
 $\zeta=-1.02:0.04:0.02$ in $[1, 1+\zeta, 0]$

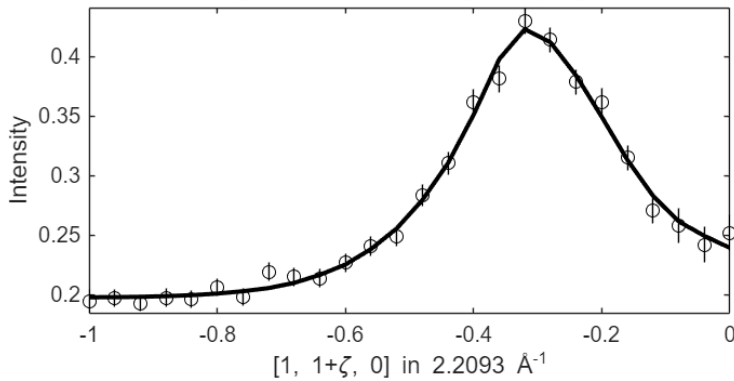


Fe_ei800, w2_800om1m1.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $98 \leq E \leq 1$
 $\zeta=-1.02:0.04:0.02$ in $[1, 1+\zeta, 0]$

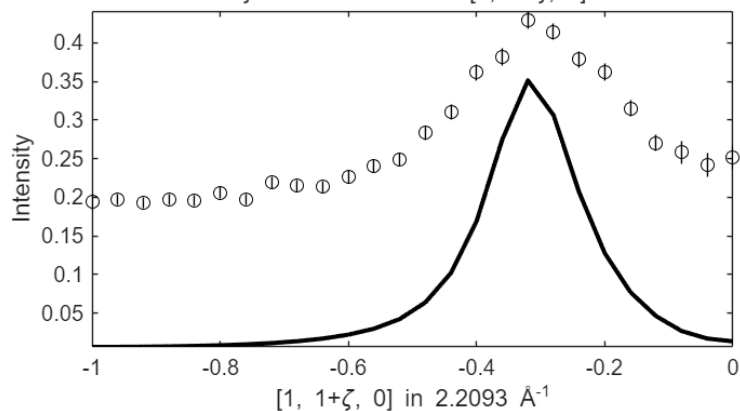


 **** En = 113±5 going to 193

Fe_ei800, w2_800om1m1.sqw
 $S=0.97\pm0.016$; $J_0=36\pm0.41$; $J_1=0\pm0$; $\gamma=71.3055\pm1.12362$
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $108 \leq E \leq$
 $\zeta=-1.02:0.04:0.02$ in $[1, 1+\zeta, 0]$

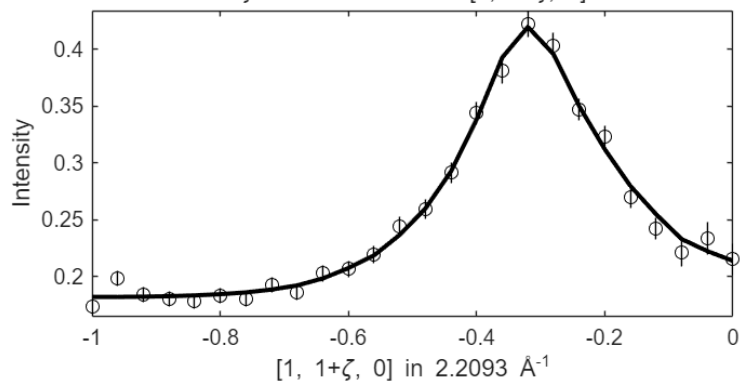


Fe_ei800, w2_800om1m1.sqw
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $108 \leq E \leq$
 $\zeta=-1.02:0.04:0.02$ in $[1, 1+\zeta, 0]$

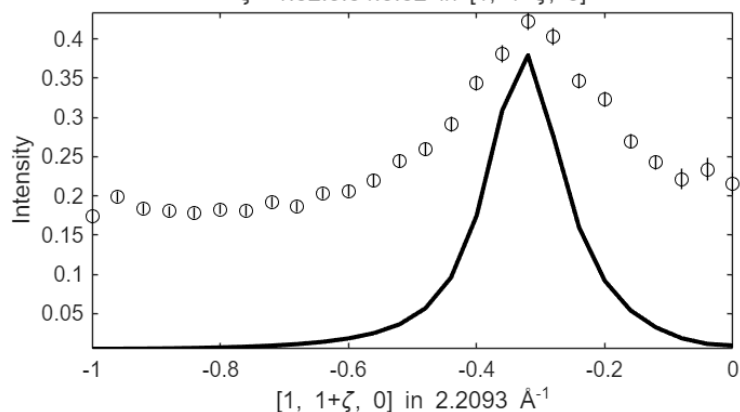


**** En = 123±5 going to 193

Fe_ei800, w2_800om1m1.sqw
 $S=0.79\pm0.014$; $J_0= 33\pm0.38$; $J_1= 0\pm 0$; $\gamma=51.9366\pm0.878482$
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $118 \leq E \leq$
 $\zeta=-1.02:0.04:0.02$ in $[1, 1+\zeta, 0]$

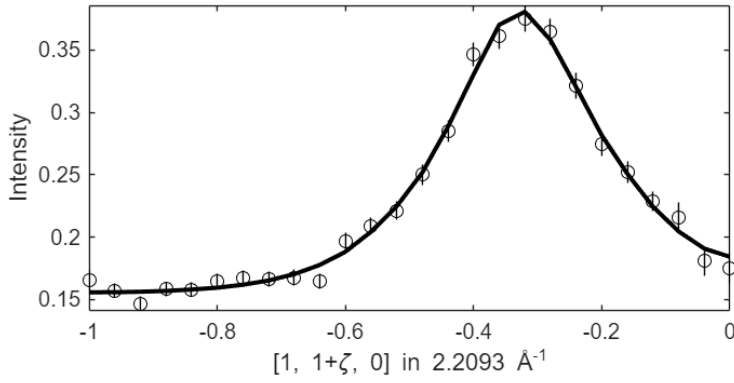


Fe_ei800, w2_800om1m1.sqw
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $118 \leq E \leq$
 $\zeta=-1.02:0.04:0.02$ in $[1, 1+\zeta, 0]$

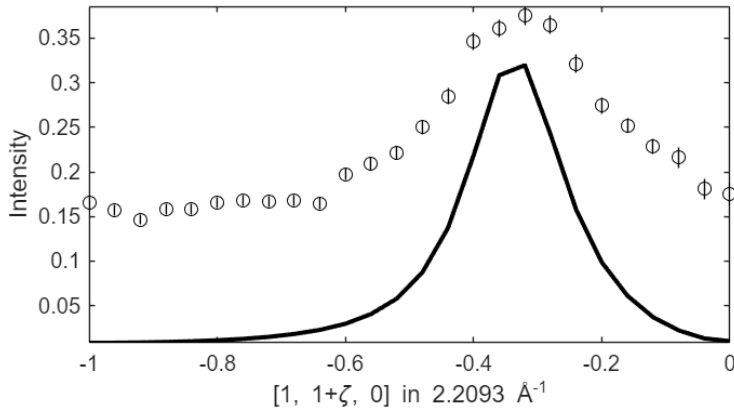


**** En = 133±5 going to 193

Fe_ei800, w2_800om1m1.sqw
 $S = 1 \pm 0.019$; $J_0 = 37 \pm 0.39$; $J_1 = 0 \pm 0$; $\gamma = 83.3473 \pm 1.27831$
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $128 \leq E \leq 128$
 $\zeta = -1.02:0.04:0.02$ in $[1, 1+\zeta, 0]$

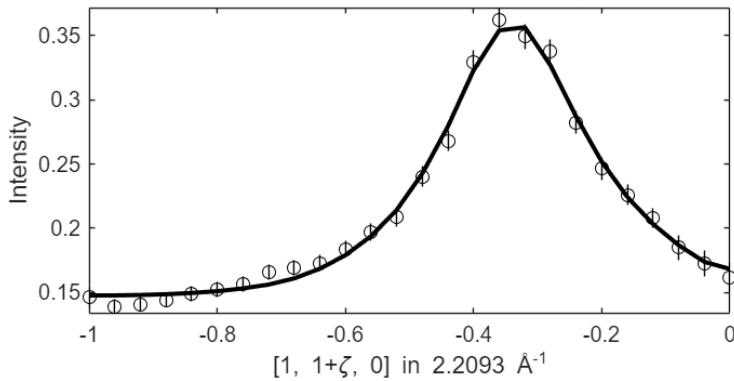


Fe_ei800, w2_800om1m1.sqw
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $128 \leq E \leq 128$
 $\zeta = -1.02:0.04:0.02$ in $[1, 1+\zeta, 0]$

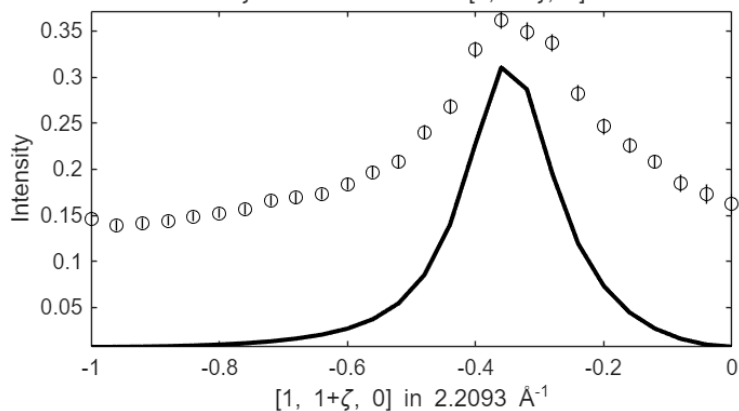


 **** En = 143±5 going to 193

Fe_ei800, w2_800om1m1.sqw
 $S = 0.86 \pm 0.073$; $J_0 = 36 \pm 1.1$; $J_1 = 0 \pm 0$; $\gamma = 71.0056 \pm 9.67281$
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $138 \leq E \leq 138$
 $\zeta = -1.02:0.04:0.02$ in $[1, 1+\zeta, 0]$

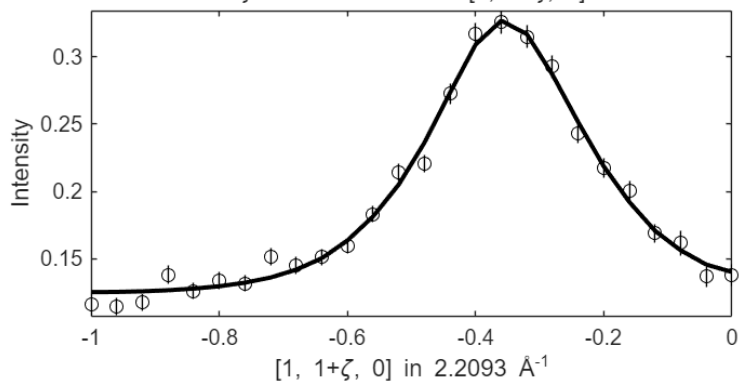


Fe_ei800, w2_800om1m1.sqw
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $138 \leq E \leq$
 $\zeta=-1.02:0.04:0.02$ in $[1, 1+\zeta, 0]$

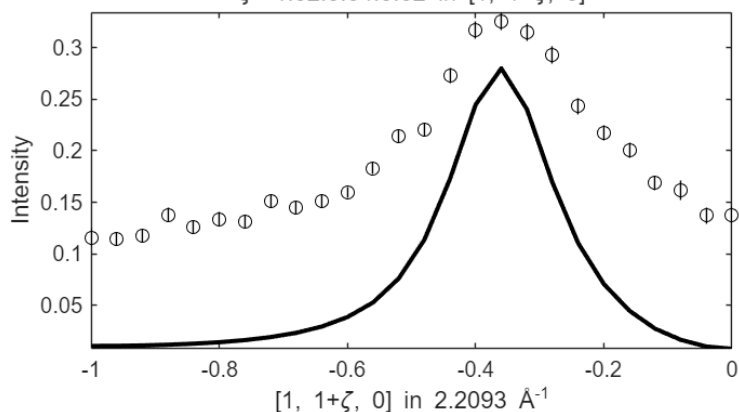


**** En = 153±5 going to 193

Fe_ei800, w2_800om1m1.sqw
 $S= 1 \pm 0.023$; $J_0= 38 \pm 0.38$; $J_1= 0 \pm 0$; $\gamma=96.7526 \pm 1.60375$
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $148 \leq E \leq$
 $\zeta=-1.02:0.04:0.02$ in $[1, 1+\zeta, 0]$

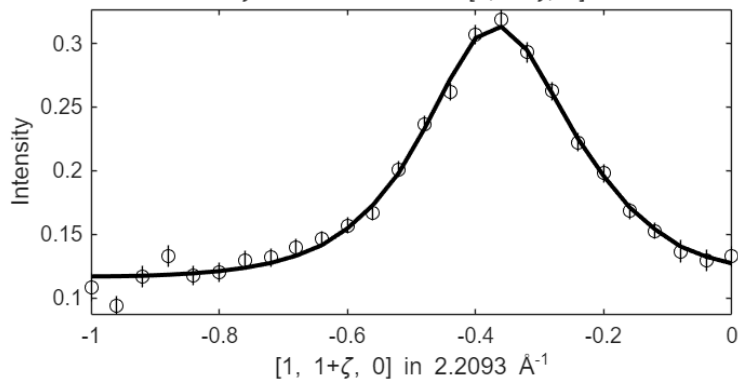


Fe_ei800, w2_800om1m1.sqw
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $148 \leq E \leq$
 $\zeta=-1.02:0.04:0.02$ in $[1, 1+\zeta, 0]$

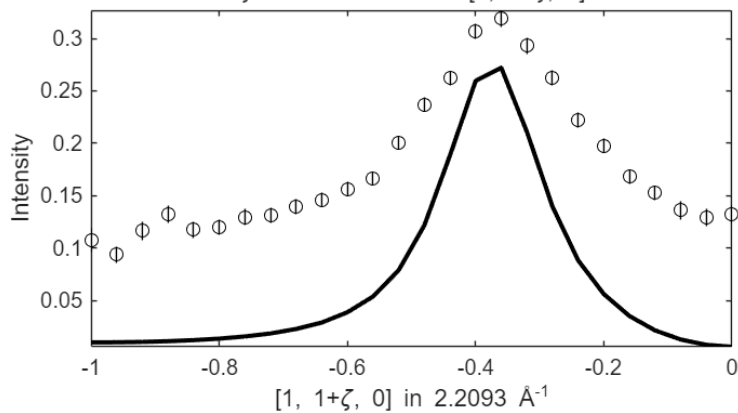


**** En = 163±5 going to 193

Fe_ei800, w2_800om1m1.sqw
 $S = 0.9 \pm 0.019$; $J_0 = 36 \pm 0.31$; $J_1 = 0 \pm 0$; $\gamma = 86.6594 \pm 1.22451$
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $158 \leq E \leq$
 $\zeta = -1.02:0.04:0.02$ in $[1, 1+\zeta, 0]$

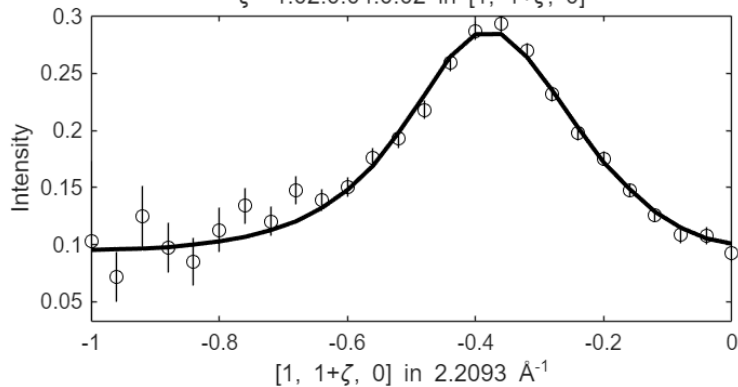


Fe_ei800, w2_800om1m1.sqw
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $158 \leq E \leq$
 $\zeta = -1.02:0.04:0.02$ in $[1, 1+\zeta, 0]$

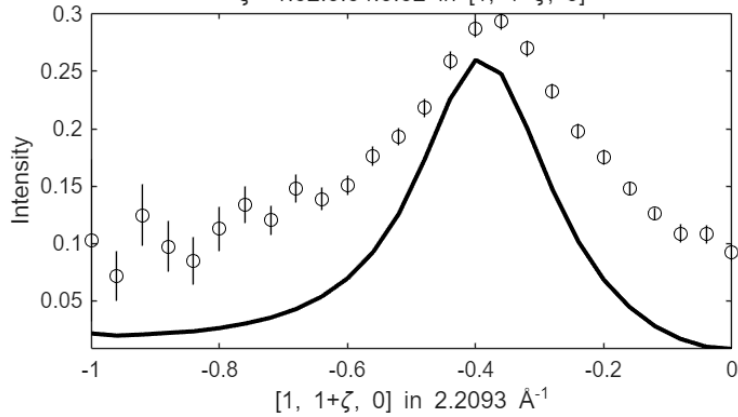


 **** En = 173±5 going to 193

Fe_ei800, w2_800om1m1.sqw
 $S = 1.4 \pm 0.044$; $J_0 = 41 \pm 0.43$; $J_1 = 0 \pm 0$; $\gamma = 152.773 \pm 2.55932$
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $168 \leq E \leq$
 $\zeta = -1.02:0.04:0.02$ in $[1, 1+\zeta, 0]$

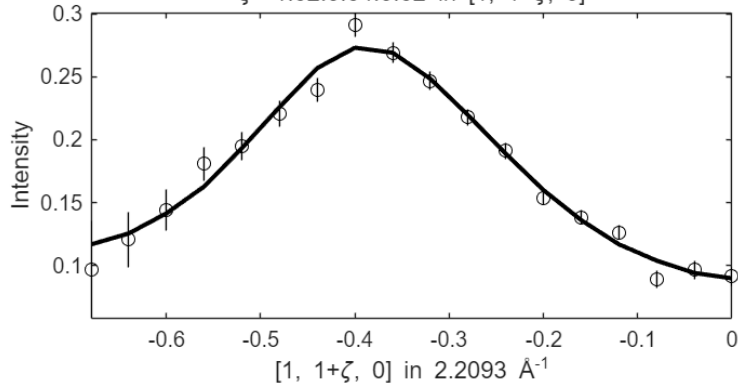


Fe_ei800, w2_800om1m1.sqw
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $168 \leq E \leq 178$
 $\zeta = -1.02:0.04:0.02$ in $[1, 1+\zeta, 0]$

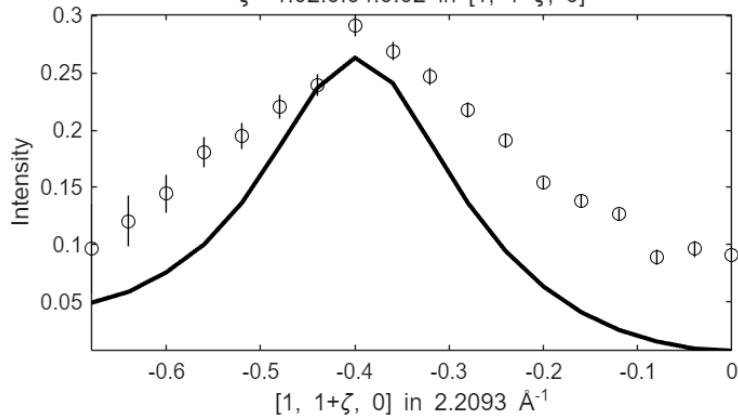


 **** En = 183±5 going to 193

Fe_ei800, w2_800om1m1.sqw
 $S = 1.4 \pm 0.062$; $J_0 = 41 \pm 0.73$; $J_1 = 0 \pm 0$; $\gamma = 150.303 \pm 3.369$
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $178 \leq E \leq 193$
 $\zeta = -1.02:0.04:0.02$ in $[1, 1+\zeta, 0]$

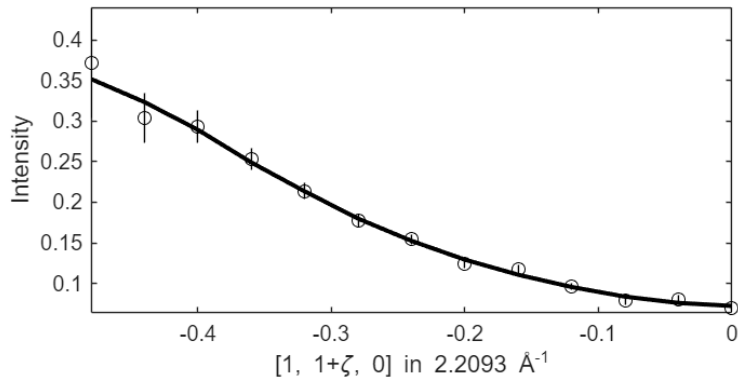


Fe_ei800, w2_800om1m1.sqw
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $178 \leq E \leq 193$
 $\zeta = -1.02:0.04:0.02$ in $[1, 1+\zeta, 0]$

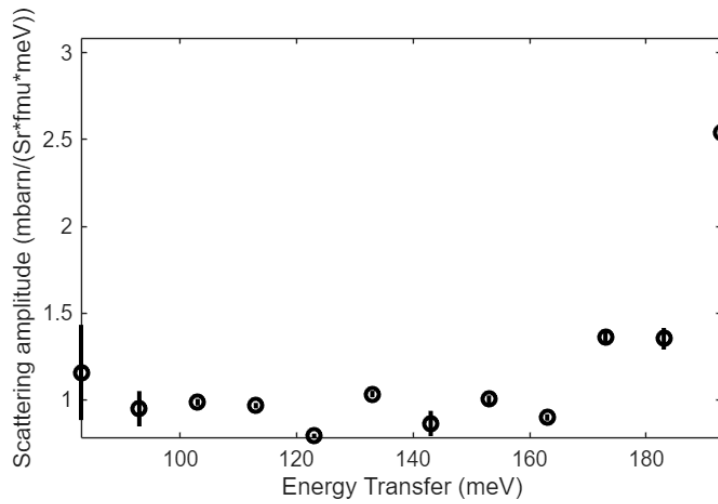
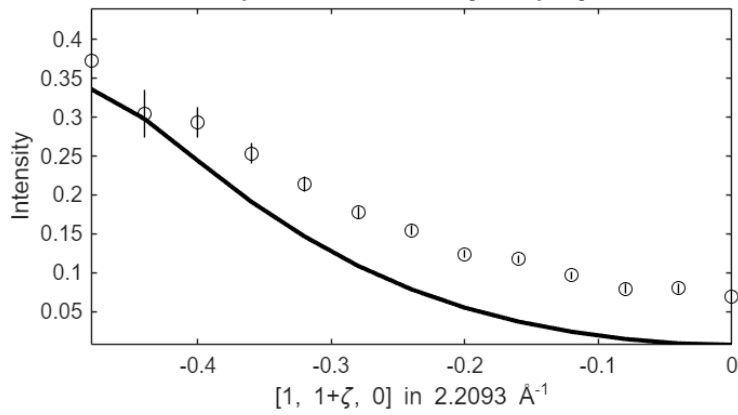


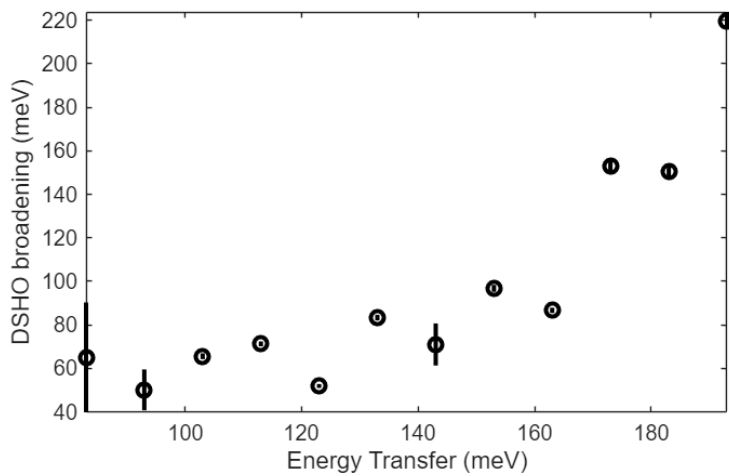
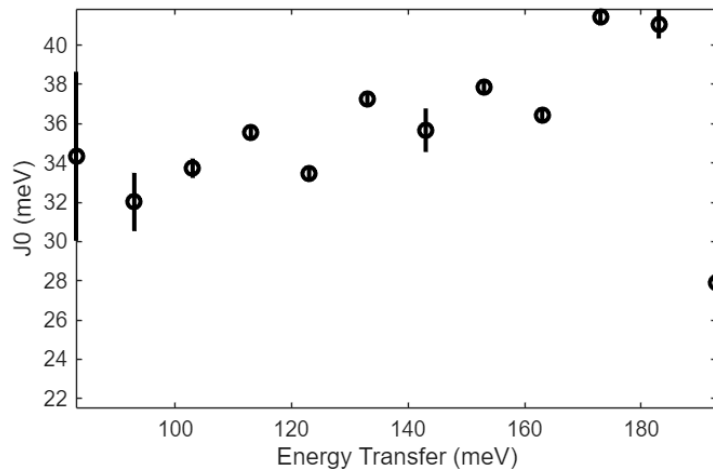
 **** En = 193±5 going to 193

Fe_ei800, w2_800om1m1.sqw
 $S = 2.5 \pm 0.55$; $J_0 = 28 \pm 6.4$; $J_1 = 0 \pm 0$; $\gamma = 219.681 \pm 4.08428$
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $188 \leq E \leq$
 $\zeta = -1.02:0.04:0.02$ in $[1, 1+\zeta, 0]$



Fe_ei800, w2_800om1m1.sqw
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $188 \leq E \leq$
 $\zeta = -1.02:0.04:0.02$ in $[1, 1+\zeta, 0]$





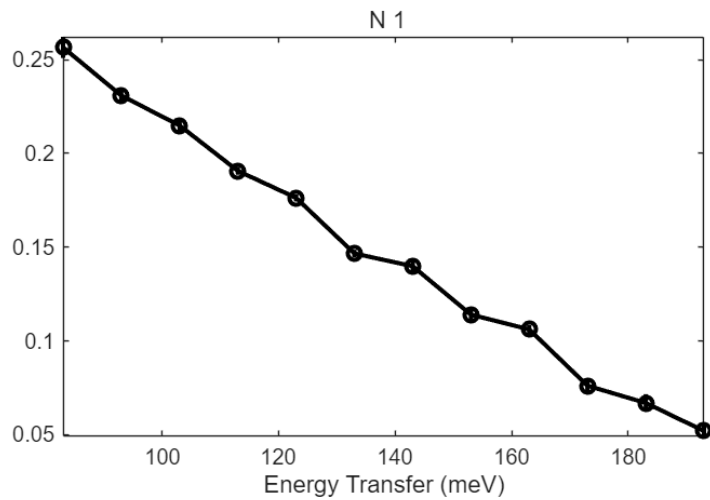
```

c1 = GH_FFCutsFits.(selectCuts);
GH_FFCutsFits.(selectCuts) = [c1{:}];

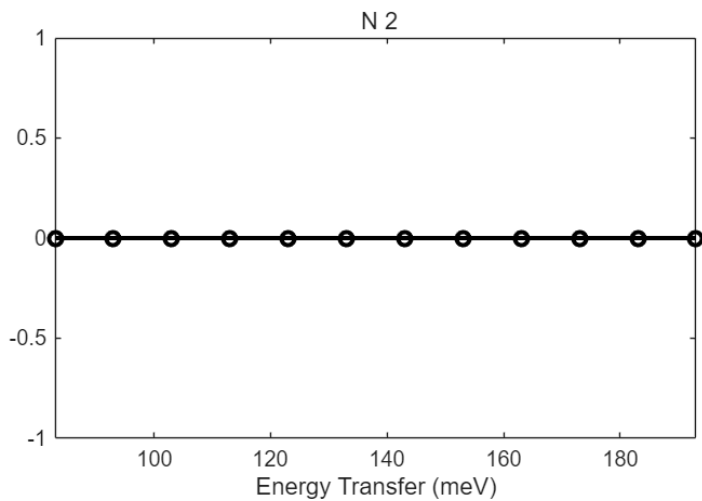
beep
beep
beep

all_fit_par = GH_FFCutsFits.([selectCuts, '_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;

```



```
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure
```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_GH.m';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_GH_',selectCuts,'.pdf']);
export(thisFile,targ_file);
```

```
fit_src_struc = GH_FFCutsFits;
save(CutFitFileGH,"fit_src_struc",-v7.3');
close all;
```

%end